

Problem Statement

Adaptive Fitness Intelligence — National AI Challenge

Prepared by Manus AI · 20 August 2026

1 The core problem

Serious recreational athletes must manually reconcile recovery data from wearables, nutrition logs and detailed strength-training records to decide whether to repeat, progress, reduce or move a session. Existing products collect portions of that picture but rarely preserve the workout-level context—sets, load, repetitions, RPE/RIR and block goal—needed to make the recommendation understandable.

1.1 Root cause, not symptom

The visible symptom is app switching: readiness in a wearable app, food in a nutrition app, and progressive-overload history in a notes file or gym tracker. The root cause is that these systems store different evidence for different purposes, making the user the integration layer. A biometrics-only score cannot determine whether yesterday's hard squat top set, a low-calorie day or a changed 8–12 week block goal is the relevant reason to progress or hold.

Physiological signals are useful but incomplete. In a 2024 study of 39 healthy adults contributing 316 paired measurements, Apple Watch HRV differed from a Polar H10/Kubios reference by -8.31 ms on average, while resting-heart-rate difference was -0.08 bpm; the authors caution against over-interpreting absolute HRV for performance monitoring. [1] A review of 90 resistance-training studies found practical links between performance and RPE/RIR, velocity and selected monitoring measures, while noting that many physiological measures are restrictive. [2]

1.2 Product implication

The first product should be an explainable, non-clinical decision-support layer. It should translate a user's existing workout notation into structured sets, surface the prior successful session as grey "ghost sets", compare within-person wearable trends against nutrition and recent training dose, and recommend a bounded next action with visible rationale. It should not diagnose fatigue, promise injury prevention or present a proprietary score as medical fact.

1.3 Why now

Sport Ireland reported that 47% of adults—1.97 million people—regularly participated in sport in 2023. [3] WHOOP offers consented access to recovery, HRV, resting heart rate, sleep, cycles and workouts, while Apple HealthKit is designed to merge permissioned multi-source health and fitness data. [4] [5] The opportunity is not to invent more signals, but to give those signals their missing training context.

2 References

1. O'Grady et al., **The Validity of Apple Watch Series 9 and Ultra 2 for Serial Measurements of Heart Rate Variability and Resting Heart Rate** (2024).
2. Helms et al., **Methods for Regulating and Monitoring Resistance Training** (2020).
3. Sport Ireland, **Irish Sports Monitor 2023**.
4. WHOOP, **Developer API Documentation**.
5. Apple, **HealthKit Documentation**.